Project Reflection Report

Lumora — Responsive Multi-Page SaaS Website

HTML5 · CSS3 · Bootstrap 5 (CDN) · Bootstrap Icons · Google Fonts · GitHub Pages

3	15+	5	10 hrs
Pages Built	Custom Sections	JS Interactions	Completion Time

Project Development Process

1. **Planning & Setup —** Created project folder with 3 HTML files, 1 shared CSS file, and a JS file.
2. **Design System First —** Defined all CSS custom properties — colors, fonts (Fraunces + DM Sans), spacing, and shadows — before writing any layout code.
3. **Shared Components —** Built the sticky navbar and footer first. Copy-pasted them across all 3 pages to keep everything consistent.
4. **Home Page (index.html) —** Built hero section with animated blobs, product mockup, logo ticker, feature cards, sign-up modal, and footer.
5. **About & Services Page (about.html) —** Built page hero, stat counters, team cards, services rows with tag chips, and 3-tier pricing cards.
6. **Contact Page (contact.html) —** Built two-column layout, live form validation, success animation, contact panel, FAQ accordion, and demo booking modal.
7. **Scroll Animations —** Added IntersectionObserver-based reveal animations across all sections — no external library used.
8. **Testing & Bug Fixes —** Tested on mobile, tablet, and desktop. Fixed navbar specificity clash and right-panel alignment issue.
9. **Deployment —** Switched all internal hrefs to relative paths and deployed via GitHub Pages from the main branch.

1. Project Overview

What is Lumora?

- Lumora is a fictional AI-powered SaaS product designed to help teams automate workflows, track analytics in real time, and collaborate securely.
- It targets businesses that want to streamline their operations using AI — positioned as a modern, enterprise-ready platform.
- The website is the product's public-facing presence: 3 pages covering the product, the team, pricing, and contact.

Project Goal

- Build a fully responsive, multi-page SaaS website using Bootstrap 5 as a base — not as a copy-paste template.
- Layer a custom design system on top of Bootstrap to achieve a polished, product-grade look.
- Consistent design tokens — CSS custom properties for all colors, shadows, and spacing.
- Font pairing: Fraunces (display headings) + DM Sans (body text).
- Color palette: cream background, lavender accent, ink dark, with blush, sage, and sky as supporting tones.
- Smooth scroll, sticky navbar, and scroll-triggered reveal animations via IntersectionObserver.

2. Page-by-Page Breakdown

Home Page — index.html

- Sticky glassmorphism navbar with responsive hamburger collapse.
- Hero section — animated blob backgrounds, product visual mockup with a live bar chart, avatar stack pill, and CTA buttons.
- Trusted-by logo ticker strip — Voltex, Prismly, Nimbus, Nexacore, Stackify.
- 4-column feature cards — AI Automation, Live Analytics, Collaboration, Enterprise Security.
- CTA banner with a modal-triggered sign-up form — Google & GitHub OAuth buttons.
- Multi-column footer with social icon buttons.

About & Services Page — about.html

- Page hero with gradient background and company mission statement.
- 4-stat counter row — number of users, automations, uptime percentage, and NPS score.
- Team section — 4 member cards with avatar initials, role, bio, and social icon links.
- Services section — 4 numbered rows (AI Automation, Analytics, Collaboration, Security) with tag chips.
- 3-tier pricing cards — Starter (free), Growth ($49/mo, featured), Enterprise (custom) with feature lists.
- CTA strip + full footer consistent with the home page.

Contact Page — contact.html

- Page hero with gradient background and quick-action buttons — Send Message & Book a Demo.
- Two-column layout — contact form on the left, contact details panel on the right.
- Form with live validation — First/Last name, Email, Company, Subject dropdown, Message textarea (600 char limit), Privacy consent checkbox.
- Success state animation on form submit, with a reset option.
- Right panel — Email, Phone, Office address, Live Chat (animated pulse badge), and social pill links.
- 'What happens next?' — a 3-step process block.
- FAQ accordion — 4 Q&A pairs with open/close animation.
- Demo booking modal triggered from the hero CTA.

3. Bootstrap 5 Components & Utilities Used

Component / Utility	Where Used
Navbar + Collapse	All 3 pages — sticky, responsive hamburger menu
Grid (rows/cols)	Every section layout — hero, features, team, pricing, contact form
Modal	Sign-up modal (Home, About) · Demo booking modal (Contact)
Utility classes	d-flex, gap-*, align-items-*, justify-content-*, mt-/mb-/px-/py-*, w-100, position-*
Form controls	Contact form — input, select, textarea, checkbox
Breakpoint classes	col-sm-*, col-md-*, col-lg-* across all pages for responsiveness

Custom additions (all hand-written, no libraries):

- Glassmorphism cards using backdrop-filter with cross-browser fallback.
- CSS variables design system for consistent theming.

- IntersectionObserver scroll-triggered reveal animations.
- Animated blob backgrounds in hero sections.
- CSS bar chart rendered entirely in the hero.
- Pulsing online badge on the Live Chat row.
- FAQ accordion with open/close animation.

4. Tools & References Used

- Bootstrap 5 Documentation — grid, utilities, components, modals, forms.
- Bootstrap Icons CDN — all icons across all pages (bi-envelope-fill, bi-stars, etc.).
- Google Fonts — Fraunces (display headings) + DM Sans (body text).
- AI assistance (Claude / ChatGPT) — used only for refining specific CSS issues such as the contact page right panel alignment and icon spacing. Page structure, design decisions, color system, and all content were done independently.
- GitHub Pages — deployment directly from the main branch of the repository.

5. Challenges & How I Solved Them

Contact page right panel alignment

- Problem: Icon + text rows were misaligned across different text lengths.
- Tried: CSS grid first — but it didn't handle varying content heights well.
- Fix: Switched to flexbox with a fixed-width icon container. This gave consistent alignment regardless of content length.

Glassmorphism cross-browser support

- Problem: backdrop-filter: blur() wasn't applying in some browsers.
- Fix: Added the -webkit-backdrop-filter prefix and set a solid fallback background so cards remained readable everywhere.

Form validation without a library

- Problem: Needed field-level validation without pulling in an external library.
- Fix: Built custom inline validation using blur event listeners per field — toggling error classes and showing/hiding messages manually.

Mobile navbar conflicting with Bootstrap

- Problem: Custom .btn-nav styles were overriding Bootstrap's .nav-link padding on mobile.
- Fix: Narrowed the CSS selectors to increase specificity without touching Bootstrap's base styles.

GitHub Pages deployment path issues

- Problem: Internal href links broke after deploying due to the repo being in a subdirectory.
- Fix: Changed all paths to relative (about.html instead of /about.html) consistently across all 3 pages.

6. Learning Journey & Completion Story

- Started with zero layout code — just a design system. Setting CSS variables and fonts first meant every page that followed felt consistent automatically.
- Building the home page first was the right call. It established the patterns for the navbar, hero, cards, and footer that carried into the other two pages.
- The about page came together faster because the base patterns were already in place. The main new challenge here was the pricing card layout and making the stat counters feel balanced.
- The contact page was the hardest — form validation logic, the two-column layout, and the right panel alignment all required multiple iterations and real problem-solving.
- Every bug (blur support, navbar clashing, GitHub paths) was fixed by understanding what was actually happening — not by guessing. That made each fix stick.
- AI tools were used only when stuck on specific CSS issues — never to generate whole sections. This kept the understanding sharp throughout.
- Finishing a 3-page, fully deployed, responsive website in 10 hours showed that starting with a solid structure and system makes everything downstream faster and cleaner.

7. Learning Outcomes

- Bootstrap 5 is a structural base, not a full design solution — custom CSS is always needed to make it look polished and unique.
- CSS custom properties make theming and cross-page consistency much easier to manage and maintain.
- CSS grid and flexbox solve different problems — knowing which to reach for in a given layout situation comes from practice, not theory.
- IntersectionObserver is a lightweight, library-free way to add effective scroll animations.
- AI tools work best as a debugging aid for specific problems — not as a shortcut for building whole structures. Using them responsibly keeps your own understanding intact.
- Multi-page project organisation — shared navbar, footer, color tokens, and typography across 3 files — builds real confidence in scalable front-end thinking.

8. Total Time Taken

Completed in **10 Hours**

- Planning & design system setup — ~1 hr
- Home page (index.html) — ~3 hrs
- About & Services page (about.html) — ~2 hrs
- Contact page (contact.html) — ~3 hrs
- Testing, fixes & GitHub Pages deployment — ~1 hr